

Measure of Dispersion/Variability

A measure of variability is one of the descriptive statistic that represents amount of dispersion in a dataset. In measure of central tendency describes the typical value, measure of variability defines how far away the data points tend to fall from the center. There are three ways to find the measure of dispersion

- (i) Range
- (ii) Quartile Deviation
- (iii) Mean Deviation
- (iv) Variance and
- (v) Standard Deviation.

Range

Range is the difference between the largest and smallest values in a dataset. It is one of the method in Measures of Dispersion/Variability.

Example: In {8,11, 5, 9, 7, 6, 2500}, the lowest value is 5 and the highest value is 2500.

So the range is $(2500 - 5) = 2495$.

Mean deviation

For data clustered near the central value, the differences of individual observations from their typical value will tend to be small. Accordingly, to obtain a measure of the total variation in the data, it is appropriate to find an average of these differences. The resulting value will be called **mean deviation**. It is also known as the **average deviation**.

Definition: The mean deviation is an average of absolute deviations of individual observations from the central value of a series.

The mean deviation is computed as the arithmetic mean of the absolute values of the deviations from the typical value of a distribution. The typical value may be the arithmetic mean, median, mode or even an arbitrary value. The median is sometimes preferred as a typical value in computing the average deviation, because the sum of the absolute values of the deviations from the median is smaller than any other value. In practice, however, the arithmetic mean is generally used. If the distribution is symmetrical, the mean is identical with the median and the average deviation is obtained.

If $x_1, x_2, \dots, x_n$ form a sample of observations, the formula for computing the mean deviation from arithmetic mean is

$$MD(\bar{x}) = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n} = \frac{\sum_{i=1}^n |d_i|}{n}$$

Where $d_i = x_i - \bar{x}$, which stands for the deviations of the individual observations from the mean; and $| |$ means that the signs of the deviations whether positive or negative, are ignored.

If a grouped frequency distribution is constructed, as is usually done with large samples, the average deviation is

$$MD(\bar{x}) = \frac{\sum_{i=1}^k f_i |x_i - \bar{x}|}{n}$$

where

$MD(\bar{x})$ = average deviation about mean

k = number of classes

x_i = mid point of the i -th class

f_i = frequency of the i -th class

Example-1: Compute mean deviation from mean of the following frequency distribution:

Weekly wage in TK	48.5-53.5	53.5-58.5	58.5-63.5	63.5-68.5	68.5-73.5	73.5-78.5	78.5-83.5	83.5-88.5	88.5-93.5	93.5-98.5	Total
Frequency	2	2	3	5	5	5	5	7	10	6	50

Solution:

Calculation table: Given mean 79.1

Class Interval	Mid-point(x_i)	Frequency(f_i)	Deviation from mean $\bar{x}$ (d_i)	$f_i d_i $
48.5-53.5	51	2	-28.1	56.2
53.5-58.5	56	2	-23.1	46.2
58.5-63.5	61	3	-18.1	54.3
63.5-68.5	66	5	-13.1	65.5
68.5-73.5	71	5	-8.1	40.5
73.5-78.5	76	5	-3.1	15.5
78.5-83.5	81	5	1.9	9.5
83.5-88.5	86	7	6.9	48.3
88.5-93.5	91	10	11.9	119
93.5-98.5	96	6	16.9	101.4
Total		50		556.4

To compute mean deviation from the arithmetic mean, the following steps may be followed:

- (i) Calculate the arithmetic mean $\bar{x}$ using the formula $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$.
- (ii) Take the absolute deviation of each mid-point from $\bar{x}$ and multiply the deviations by the corresponding frequency to get $f_i |x_i - \bar{x}|$.
- (iii) Sum these deviations as before and divide this sum by the total frequency.
- (iv) The resulting value in step (iii) is the mean deviation.

The computed values of the mean $\bar{x}$ and $MD(\bar{x})$ appear below:

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = 79.1(\text{given})$$

$$\begin{aligned} MD(\bar{x}) &= \frac{\sum_{i=1}^k f_i |x_i - \bar{x}|}{n} \\ \Rightarrow MD(\bar{x}) &= \frac{\sum_{i=1}^k f_i |d_i|}{n} \\ &= \frac{556.4}{50} \\ &= 11.1 \end{aligned}$$

Mean Deviation can also be determined by formula given below:

Mean Deviation from arithmetic mean ($\bar{x}$)

$$MD(\bar{x}) = \frac{1}{n} \left[\sum_{x_i > \bar{x}} f_i x_i - \sum_{x_i < \bar{x}} f_i x_i + \bar{x} \left(\sum_{x_i < \bar{x}} f_i - \sum_{x_i > \bar{x}} f_i \right) \right]$$

Mean Deviation from Median (Me)

$$MD(Me) = \frac{1}{n} \left[\sum_{x_i > Me} f_i x_i - \sum_{x_i < Me} f_i x_i + Me \left(\sum_{x_i < Me} f_i - \sum_{x_i > Me} f_i \right) \right]$$

Mean Deviation from Mode (Mo)

$$MD(Mo) = \frac{1}{n} \left[\sum_{x_i > Mo} f_i x_i - \sum_{x_i < Mo} f_i x_i + Mo \left(\sum_{x_i < Mo} f_i - \sum_{x_i > Mo} f_i \right) \right]$$

Example-2: Use the data in example-1 to compute the mean deviation from arithmetic mean (79.1), from median (81.2), from mode (90.8) of the distribution.

Solution:

Class Interval	Mid-point(x_i)	Frequency(f_i)	$f_i x_i$
48.5-53.5	51	2	102
53.5-58.5	56	2	112
58.5-63.5	61	3	183
63.5-68.5	66	5	330
68.5-73.5	71	5	355
73.5-78.5	76	5	380
78.5-83.5	81	5	405
83.5-88.5	86	7	602
88.5-93.5	91	10	910
93.5-98.5	96	6	576
Total		50	3955

Mean Deviation from arithmetic mean:

Mean of this distribution is 79.1 (given), which lies between 76 and 81.

Hence

$$\sum_{x_i > \bar{x}} f_i x_i = 405 + 602 + 910 + 576 = 2493$$

$$\sum_{x_i < \bar{x}} f_i x_i = 102 + 112 + 183 + 330 + 355 + 380 = 1462$$

$$\sum_{x_i < \bar{x}} f_i = 2 + 2 + 3 + 5 + 5 + 5 = 22$$

$$\sum_{x_i > \bar{x}} f_i = 5 + 7 + 10 + 6 = 28$$

Then we obtain from the formula

$$\begin{aligned}
 MD(\bar{x}) &= \frac{1}{n} \left[\sum_{x_i > \bar{x}} f_i x_i - \sum_{x_i < \bar{x}} f_i x_i + \bar{x} \left(\sum_{x_i < \bar{x}} f_i - \sum_{x_i > \bar{x}} f_i \right) \right] \\
 \Rightarrow MD(\bar{x}) &= \frac{1}{50} [2493 - 1462 + 79.1(22 - 28)] \\
 &= 11.13
 \end{aligned}$$

Mean Deviation from median:

Median of this distribution is 81.2 (given), which lies between 81 and 86.

Hence

$$\begin{aligned}\sum_{x_i > Me} f_i x_i &= 602 + 910 + 576 = 2088 \\ \sum_{x_i < Me} f_i x_i &= 102 + 112 + 183 + 330 + 355 + 380 + 405 = 1867 \\ \sum_{x_i < Me} f_i &= 2 + 2 + 3 + 5 + 5 + 5 + 5 = 27 \\ \sum_{x_i > Me} f_i &= 7 + 10 + 6 = 23\end{aligned}$$

Then we obtain from the formula

$$\begin{aligned}MD(Me) &= \frac{1}{n} \left[\sum_{x_i > Me} f_i x_i - \sum_{x_i < Me} f_i x_i + Me \left(\sum_{x_i < Me} f_i - \sum_{x_i > Me} f_i \right) \right] \\ \Rightarrow MD(Me) &= \frac{1}{50} [2088 - 1867 + 81.2(27 - 23)] \\ &= 10.9\end{aligned}$$

Mean Deviation from mode:

Mode of this distribution is 90.8 (given), which lies between 86 and 91.

Hence

$$\begin{aligned}\sum_{x_i > Mo} f_i x_i &= 910 + 576 = 1486 \\ \sum_{x_i < Mo} f_i x_i &= 102 + 112 + 183 + 330 + 355 + 380 + 405 + 602 = 2469 \\ \sum_{x_i < Mo} f_i &= 2 + 2 + 3 + 5 + 5 + 5 + 5 + 7 = 34 \\ \sum_{x_i > Mo} f_i &= 10 + 6 = 16\end{aligned}$$

Then we obtain from the formula

$$\begin{aligned}MD(Mo) &= \frac{1}{n} \left[\sum_{x_i > Mo} f_i x_i - \sum_{x_i < Mo} f_i x_i + Mo \left(\sum_{x_i < Mo} f_i - \sum_{x_i > Mo} f_i \right) \right] \\ \Rightarrow MD(Mo) &= \frac{1}{50} [1486 - 2469 + 90.8(34 - 16)] \\ &= 13.03\end{aligned}$$

Variance and Standard Deviation

In statistics, variance and standard deviation play a vital role in measurement. The measurement of how data points vary from their mean value is known as variance and the measure of the distribution of the statistical data is called the standard deviation. The main difference between standard difference and variance is that the standard deviation is represented with the same units as the mean data whereas the variance is represented in terms of squared units.

Variance

The measurement of the distance between the mean or average value of a data set and how far a data point has dispersed is called the variance. It is represented as " σ^2 ".

Properties of Variance

1. Since the sum of variance is positive because each term of the variance is always going to be squared, the result of the variance is always going to be either a positive number or it is going to equal to zero.
2. The unit of variance is always going to be squared. For example, the variance of the volume of different vessels that are estimated in liters will be given as liter square.

Standard Deviation

The standard deviation measures the spread of the statistical data. The deviation of data from the average position or its mean value is measured by distribution. With the help of the method that is used for the deviation of data points, the degree of dispersion is computed. The symbol " σ " represented the standard deviation.

Properties of Standard Deviation

1. The square root of the means of all the squares of all values in a data set is described by the standard deviation. In other terms, the standard deviation is also called the root mean square deviation.
2. The smallest value of the standard deviation can only be number zero. This is because the value of the standard deviation cannot be a negative number.
3. The standard deviation's value is low or close to zero when the data sets are similar to each other. But if the value of these data sets has a huge difference or have a high differential value, then the value of the standard deviation is going to be high.

Variance and Standard Deviation Formula

As explained earlier, the measurement of the distance between the mean or average value of a data set and how far a data point has dispersed is called the variance. The standard deviation measures the spread of the statistical data. The deviation of data from the average position or its mean value is measured by distribution. With the help of the method that is used for the deviation of data points, the degree of dispersion is computed. Let's consider an example. Let's say a set has 'n' number of elements and the elements in the set are $\{x_1, x_2, x_3, x_4, \dots, x_n\}$. Then, the formulas for standard deviation variance formula goes by:

$$\text{Variance, } \sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}.$$

$$\text{Standard deviation, } = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}.$$

where x_i is the data set values and $\bar{x}$ is the mean of the data set.

Variance and standard deviation relationship

Variance is equal to the average squared deviations from the mean, while standard deviation is the number's square root. Also, the standard deviation is a square root of variance. Both measures exhibit variability in distribution, but their units vary: Standard deviation is expressed in the same units as the original values, whereas the variance is expressed in squared units.

Example-3: If a die is rolled, then find the variance and standard deviation of the possibilities.

Solution: When a die is rolled, the possible outcome will be 6. So the sample space, $n = 6$ and the data set = $\{1;2;3;4;5;6\}$.

To find the variance, we have, $\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$.

Now the mean of the data set, $\bar{x} = \frac{1+2+3+4+5+6}{6} = 3.5$

and

x_i	$\bar{x}$	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
1	3.5	-2.5	6.25
2		-1.5	2.25
3		-0.5	0.25

x_i		$x_i - \bar{x}$	$(x_i - \bar{x})^2$
4		0.5	0.25
5		1.5	2.25
6		2.5	6.25
Total			$\sum_{i=1}^n (x_i - \bar{x})^2 = 17.5$

So the variance is $\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$
 $= \frac{17.5}{6} = 2.917.$

Now, the standard deviation, $\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$
 $= \sqrt{2.917} = 1.708.$

Example-4: You flip a coin 100 times and it lands on heads 44 times. You then use the same coin and do another 100 flips. This time it lands on heads 49 times. You repeat this experiment a total of 10 times and get the following results for the number of heads {44, 49, 52, 62, 53, 48, 54, 49, 46, 51}. Compute the mean, variance and standard deviation of this data set.

Solution: Given the data set, {44, 49, 52, 62, 53, 48, 54, 49, 46, 51}. The sample size is $n=10$.

In order to find the mean, $\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{44+49+52+62+53+48+54+49+46+51}{10} = \frac{508}{10} = 50.8$

To find the variance, we have, $\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$

and

x_i	$\bar{x}$	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
44	50.8	-6.8	46.24
49		-1.8	3.24
52		1.2	1.44
62		11.2	125.44
53		2.2	4.84
48		-2.8	7.84

x_i		$x_i - \bar{x}$	$(x_i - \bar{x})^2$
54		3.2	10.24
49		-1.8	3.24
46		-4.8	23.04
51		0.2	0.04
Total			$\sum_{i=1}^n (x_i - \bar{x})^2 = 225.6$

So the variance is $\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$
 $= \frac{225.6}{10} = 22.56.$

Now, the standard deviation, $\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$
 $= \sqrt{22.56} = 4.749.$

Population Variance for group data: $\sigma^2 = \frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n}$

Sample Variance for group data: $s^2 = \frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n-1}$

Another formula for easier calculation: $\sigma^2 = \frac{n \sum f_i x_i^2 - (\sum f_i x_i)^2}{n(n-1)}$

Example-5: Computation of sample variance and standard deviation for group data:

x_i	f_i	$x_i f_i$	$\bar{x}$	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$f_i (x_i - \bar{x})^2$
3	2	6	6	-3	9	18
5	3	15		-1	1	3
7	2	14		1	1	2
8	2	16		2	4	8
9	1	9		3	9	9
Total	10	60				40

$$\bar{x} = \frac{\sum x_i f_i}{n} = \frac{60}{10} = 6$$

$$\text{Sample Variance: } s^2 = \frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n-1}$$

$$\begin{aligned} \Rightarrow s^2 &= \frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n-1} \\ &= \frac{40}{9} \\ &= 4.44 \end{aligned}$$

$$\begin{aligned} \text{Standard deviation: } SD &= \sqrt{4.44} \\ &= 2.1 \end{aligned}$$

Example-6: The lengths of 32 leaves were measured correct to the nearest *mm*. Find mean length and the standard deviation.

Length	20-22	23-25	26-28	29-31	32-34
Frequency	3	6	12	9	2

Solution:

Length	x_i	x_i^2	f_i	$f_i x_i$	$f_i x_i^2$
20-22	21	441	3	63	1323
23-25	24	576	6	144	3456
26-28	27	729	12	324	8748
29-31	30	900	9	270	8100
32-34	33	1083	2	66	2178
Total			32	867	23805

Here

$$\bar{x} = \frac{\sum x_i f_i}{n} = \frac{867}{32} = 27.1$$

Then,

$$\begin{aligned}\sigma^2 &= \frac{n \sum f_i x_i^2 - (\sum f_i x_i)^2}{n(n-1)} \\ &= \frac{32(23805) - (867)^2}{32 \times 31} \\ &= 10.1\end{aligned}$$

Now,

$$\begin{aligned}SD &= \sqrt{10.1} \\ &= 3.17\end{aligned}$$

Hence mean 27.1 mm, variance 10.1 mm^2 , standard deviation 3.17 mm.

Example-7: Find the mean deviation from the mean and variance of the series $a, a + d, a + 2d, \dots, a + 2nd$.

Solution: The series consists of $2n + 1$ terms. Hence the mean of the series is

$$\begin{aligned}\bar{x} &= \frac{a + (a + d) + (a + 2d) + \dots + (a + 2nd)}{2n + 1} \\ &= \frac{(2n + 1)a + d(1 + 2 + 3 + \dots + 2n)}{2n + 1} \\ &= \frac{(2n + 1)a + 2nd \frac{(2n + 1)}{2}}{2n + 1} \\ &= a + nd\end{aligned}$$

The mean deviation from the mean is

$$MD(\bar{x}) = \frac{1}{2n + 1} \sum |x_i - \bar{x}|$$

Substituting $a, a + d, a + 2d, \dots, a + 2nd$ successively for x_i and $a + nd$ for $\bar{x}$, we have

$$\begin{aligned}MD(\bar{x}) &= \frac{1}{2n + 1} \sum |x_i - \bar{x}| \\ &= \frac{1}{2n + 1} [nd + (n - 1)d + \dots + 2d + d + 0 + d + 2d + \dots + (n - 1)d + nd] \\ &= \frac{2d}{2n + 1} (1 + 2 + \dots + n) \\ &= \frac{nd(n + 1)}{2n + 1}\end{aligned}$$

The variance of the series is

$$\begin{aligned}s_x^2 &= \frac{1}{2n + 1} \sum |x_i - \bar{x}|^2 \\ &= \frac{1}{2n + 1} [n^2 d^2 + (n - 1)^2 d^2 + \dots + d^2 + 0 + d^2 + \dots + (n - 1)^2 d^2 + n^2 d^2]\end{aligned}$$

$$\begin{aligned}
&= \frac{2d^2}{2n+1} [1^2 + 2^2 + \cdots \dots \dots + n^2] \\
&= \frac{2d^2}{2n+1} \left[\frac{n(n+1)(2n+1)}{6} \right] \\
&= \frac{nd^2(n+1)}{3}
\end{aligned}$$

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